

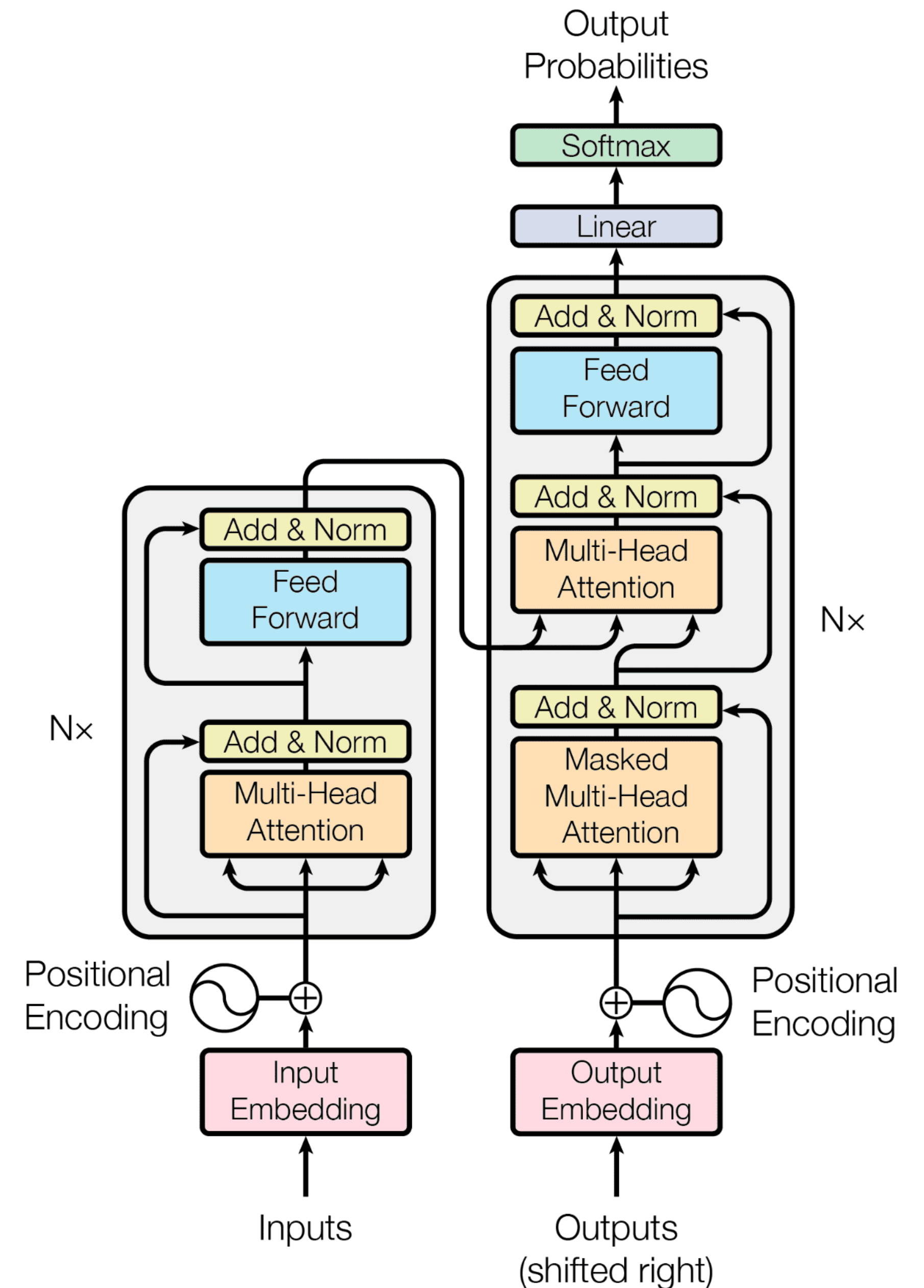
BERT and GPT

Agenda

- **BERT**
 - And its modifications
- **GPT**
 - And why it is so good
- Tricks for training Transformer models

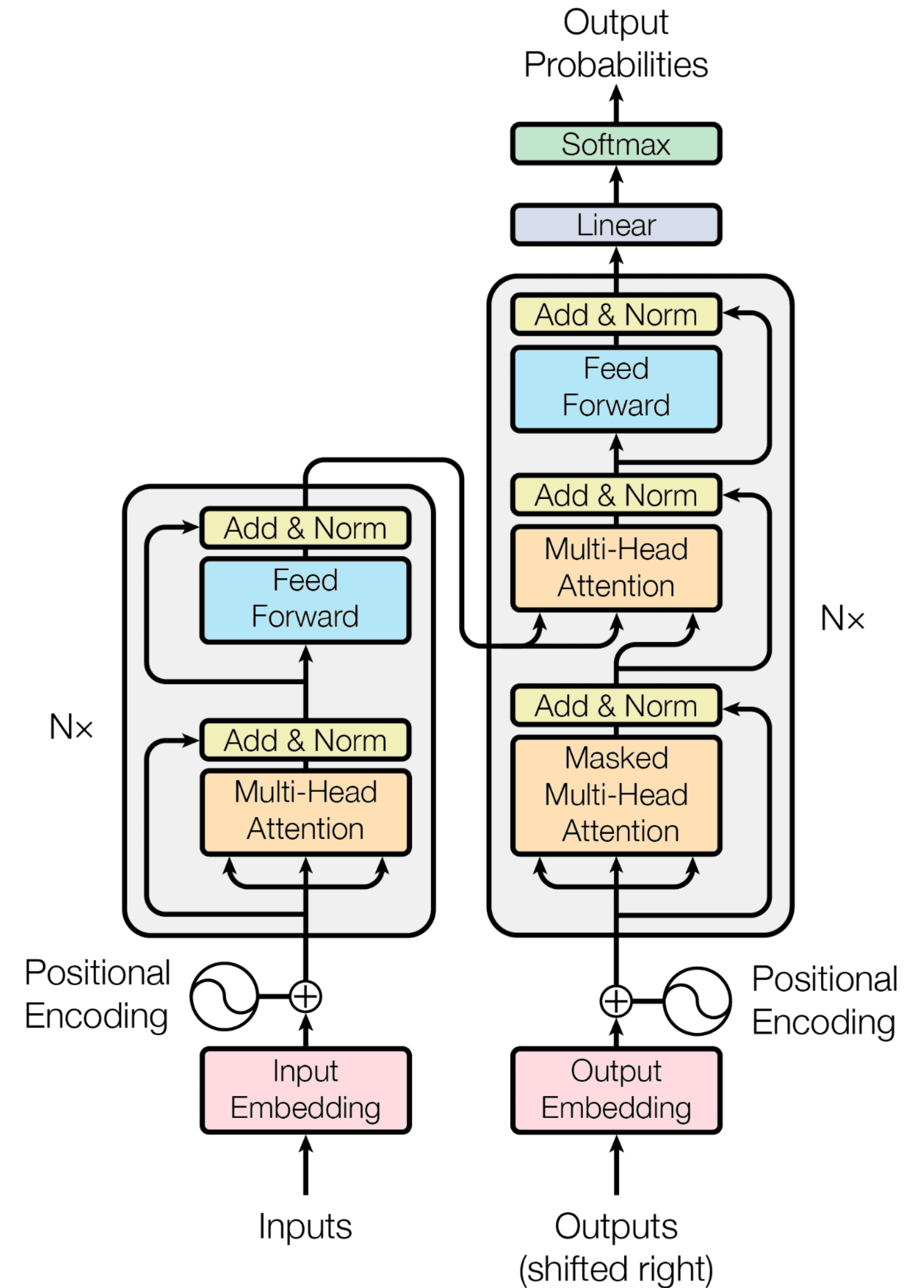
Transformer

- A Seq2seq model consisting of **Encoder** and **Decoder**
- **Encoder** extracts information from the text
- **Decoder** generates new text based on this information
- Works well thanks to the attention mechanism



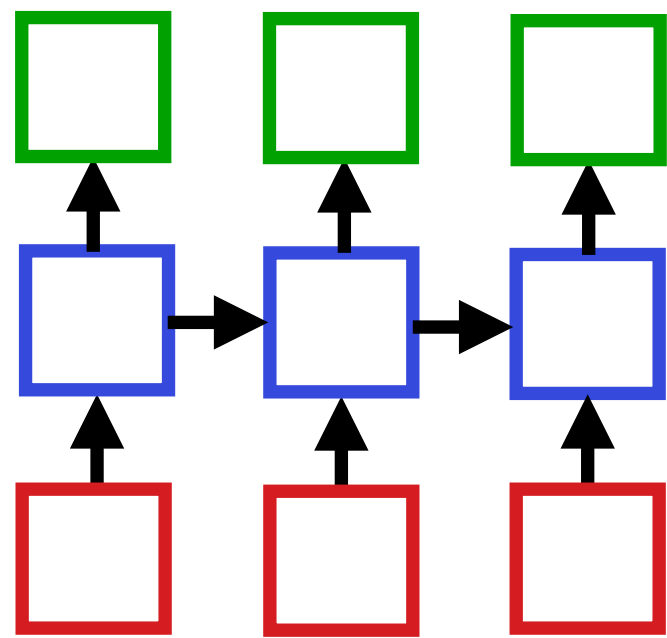
Transformer

Can we adapt this model to other types of tasks?



What types of tasks are there?

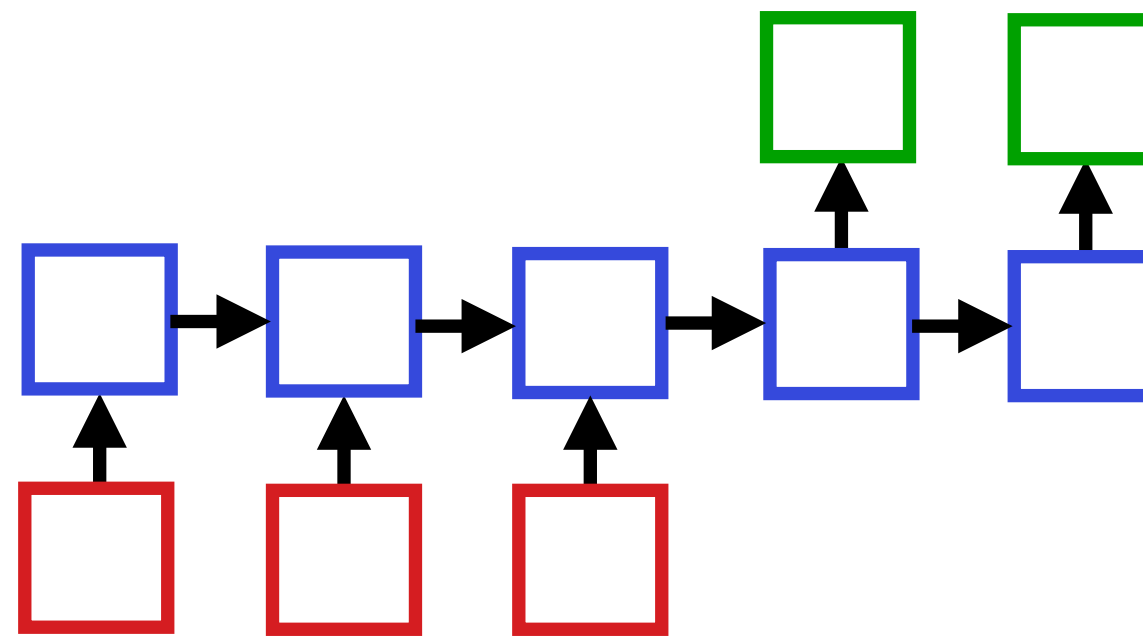
many-to-many
same length



Token classification:

- Named Entity Recognition
- Part-of-speech tagging

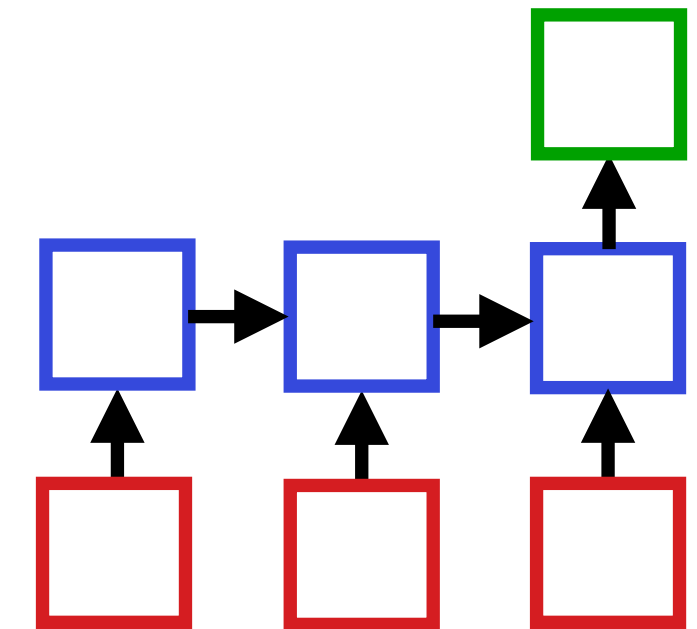
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different lengths



Seq2seq:

- Machine translation
- Summarization
- Style transfer

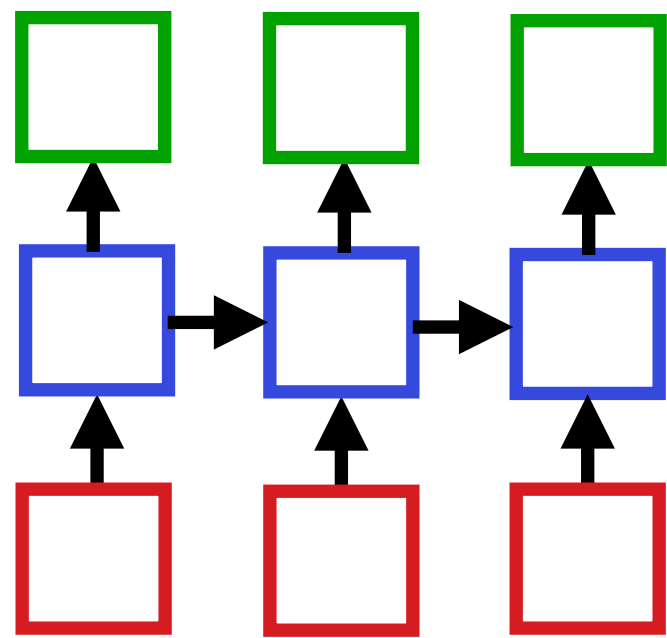
many-to-one



- Classification
- Regression

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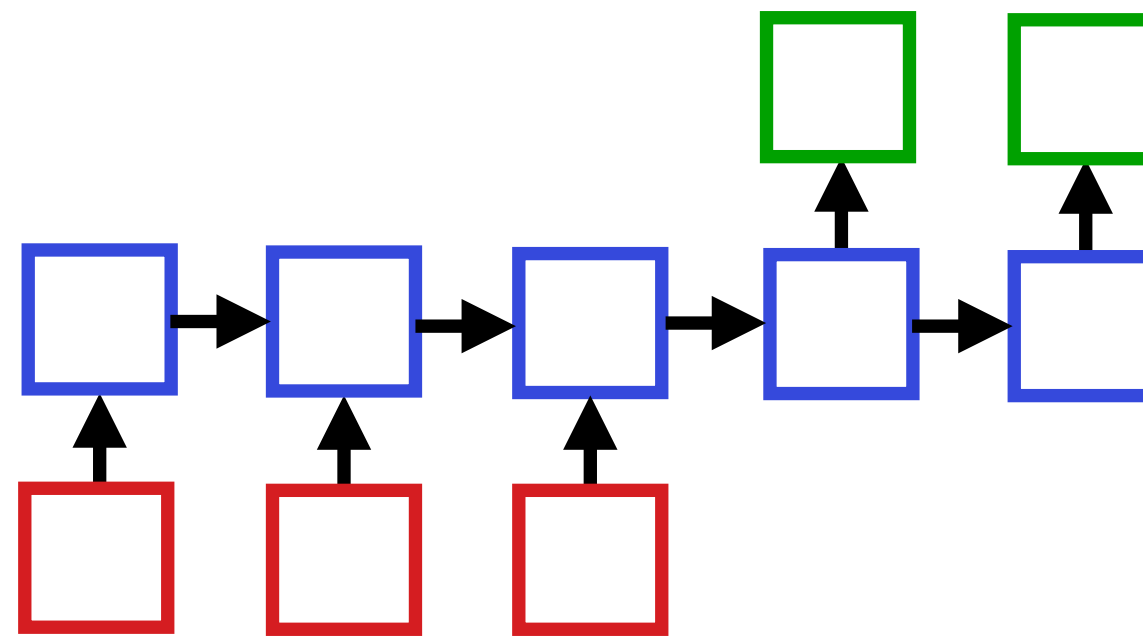
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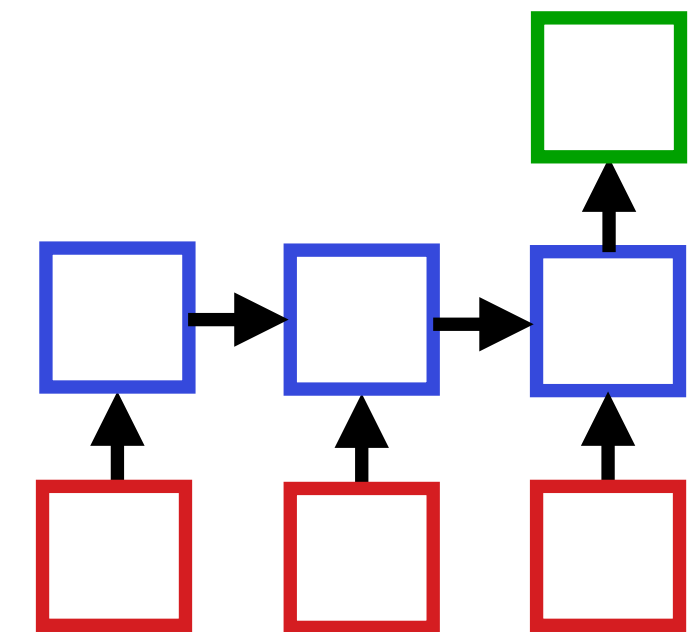
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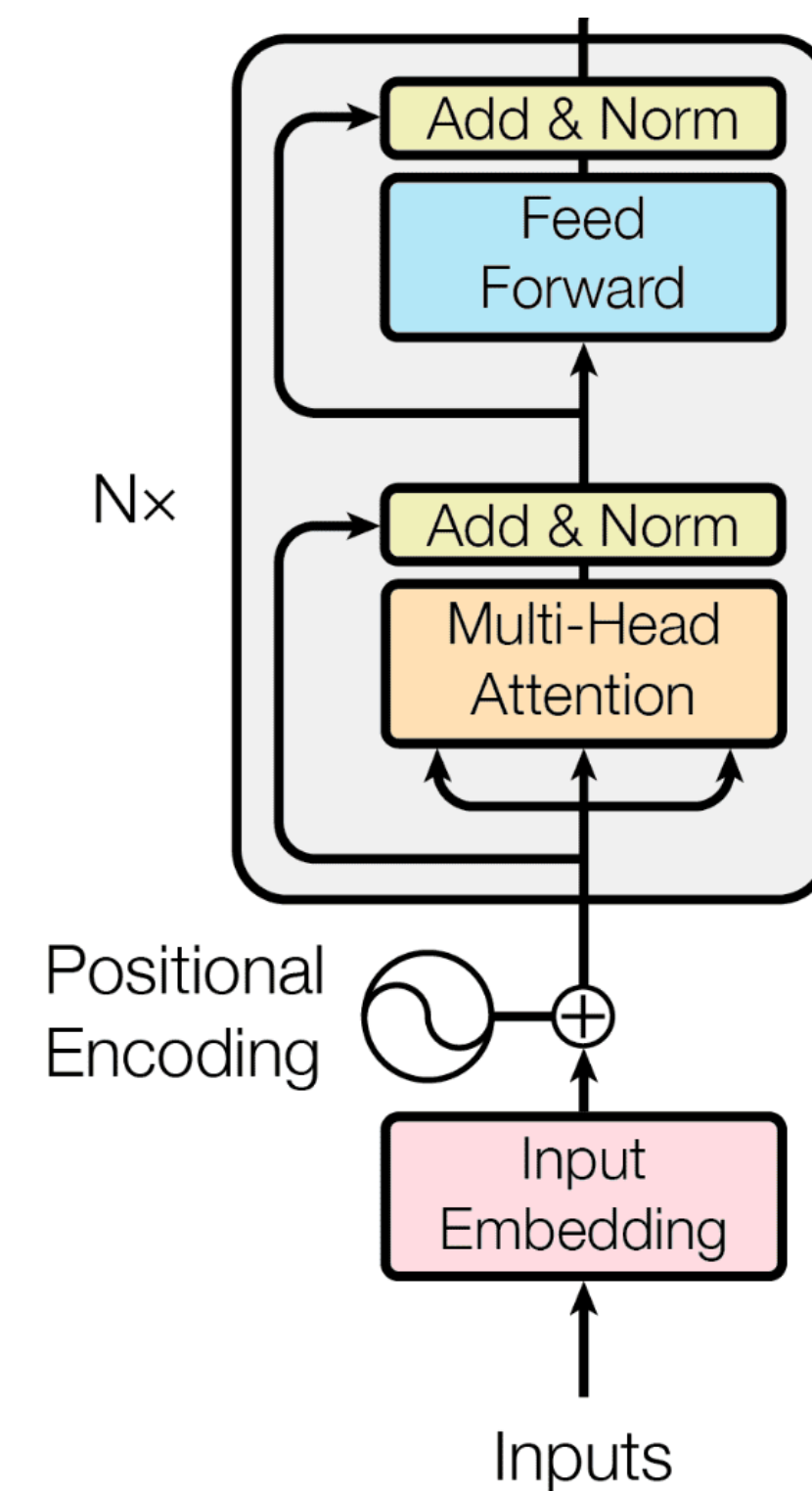
No need to generate text
We can use only **Encoder**!

Need to generate text

BERT (2018)

Bidirectional Encoder Representations

- A model based on the Transformer **Encoder**
- It is primarily used to solve arbitrary classification problems.
- The most important thing is the BERT pre-training method.



BERT: Motivation

- Transformer models can be huge.
- The base BERT contains 110M parameters.
- Training such a model requires a lot of data.
- In classification tasks, labeled data is usually insufficient.

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Let's **pre-train** a model to understand language using **unlabeled** data!

BERT: Training

To pretrain any model without labeling, it is necessary to create an artificial task.

BERT pretrains by simultaneously solving two tasks:

- Masked Language Modeling (**MLM**)
- Next Sentence Prediction (**NSP**)

Masked Language Modelling (MLM)

Idea: We mask several random tokens and ask the model to reconstruct them. The model learns to understand the meaning of words and how they are related.

Algorithm:

Randomly select 15% of the tokens. Of these,

- 80% are masked
- 10% are replaced with random tokens (to prevent the model from overfitting to [MASK])
- 10% are left unchanged (to prevent the model from considering all tokens incorrect)

Compute loss for all selected tokens

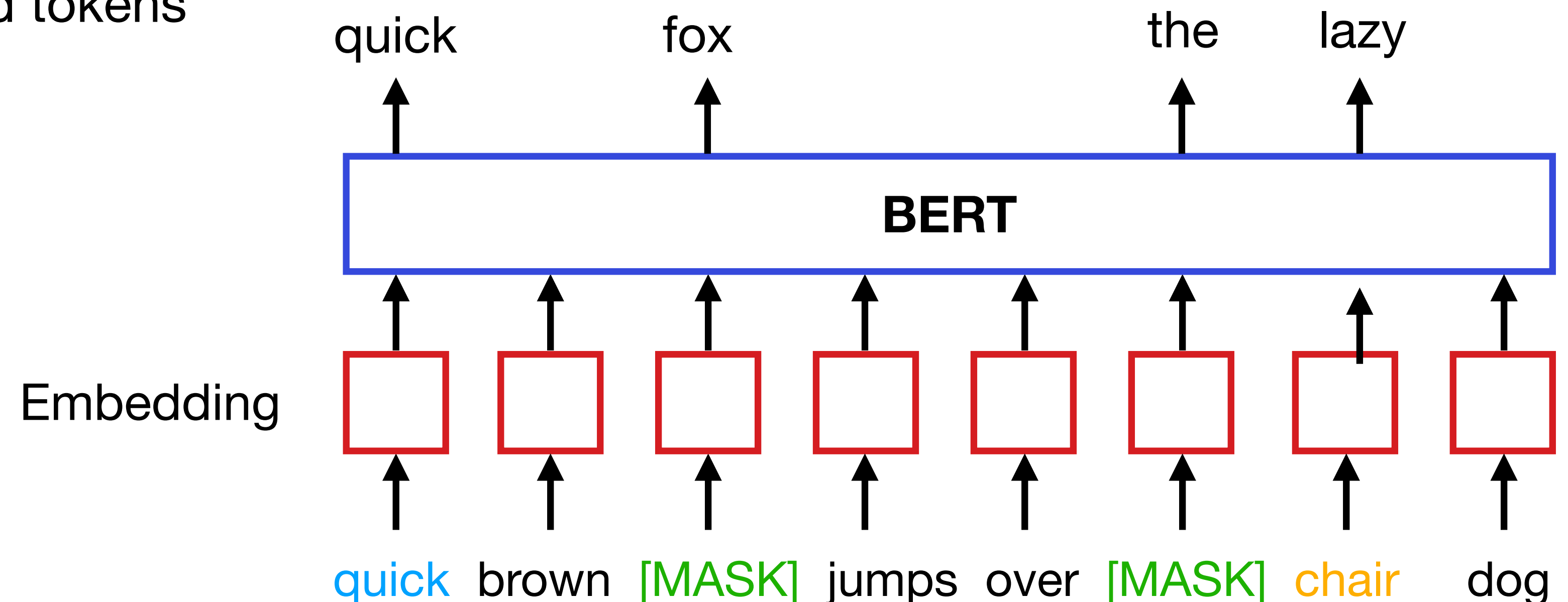
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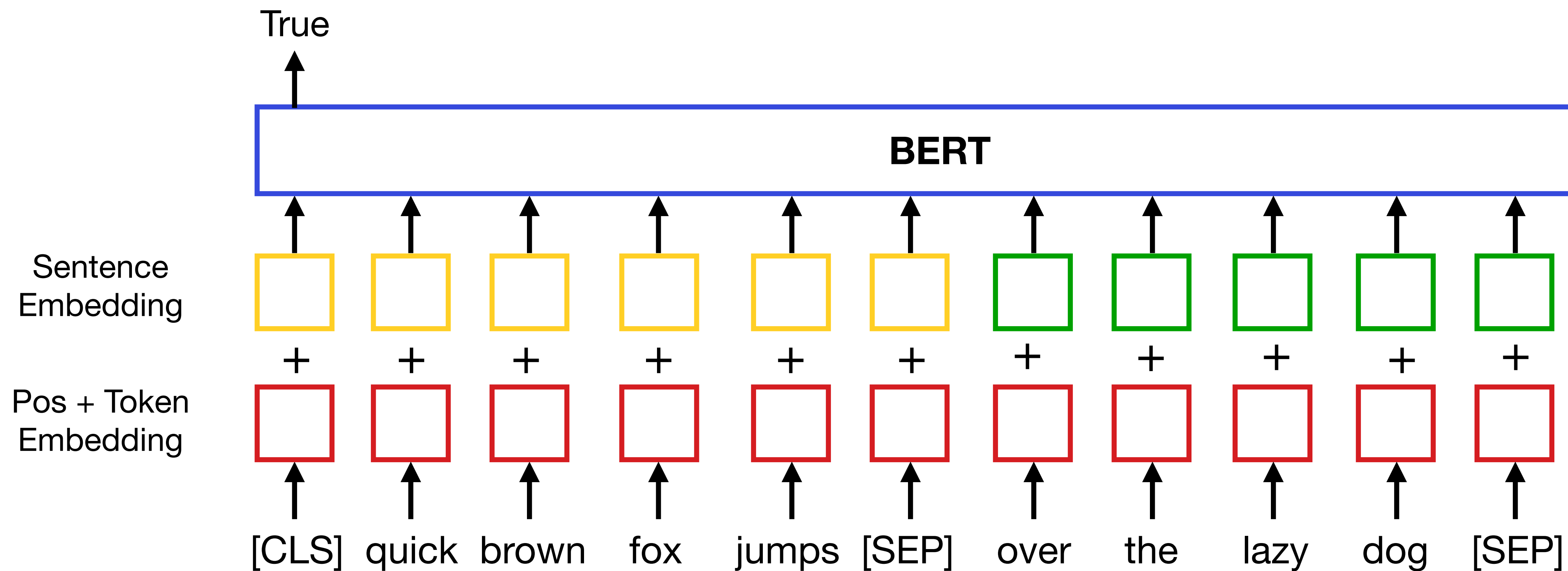


Next Sentence Prediction (NSP)

Idea: We want to teach a model to extract information from a text as a whole. This is useful for text classification.

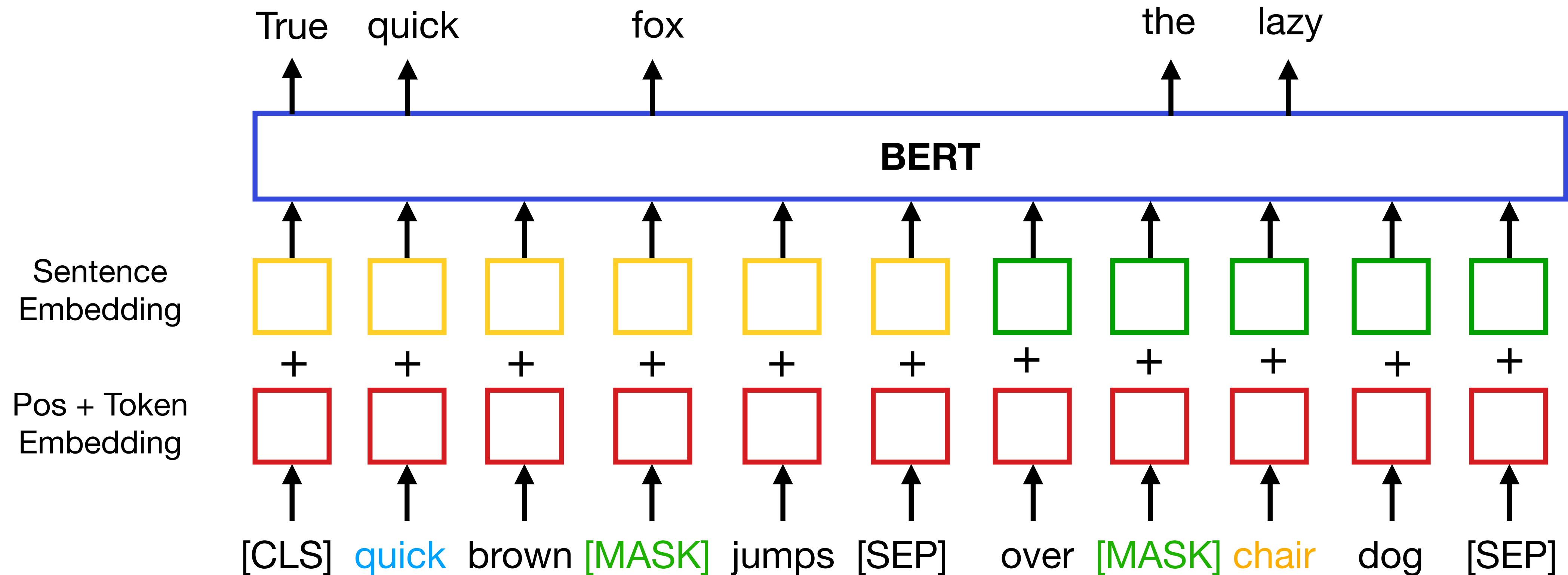
- We feed two concatenated texts into BERT.
- In half the cases, the second text is a continuation of the first, and in the other half, it is random.
- We add the [CLS] token to the beginning and [SEP] to the end of each text.
- The model's task is to predict whether the texts are related based on the [CLS] vector representation.

Next Sentence Prediction (NSP)



Combination of two tasks

BERT learns to solve two problems simultaneously.



BERT: Results

- Using pretraining allowed the model to understand text well.
- At the time of its creation, BERT outperformed its closest competitor by 7% on the GLEU benchmark.

Datasets (16 GB):

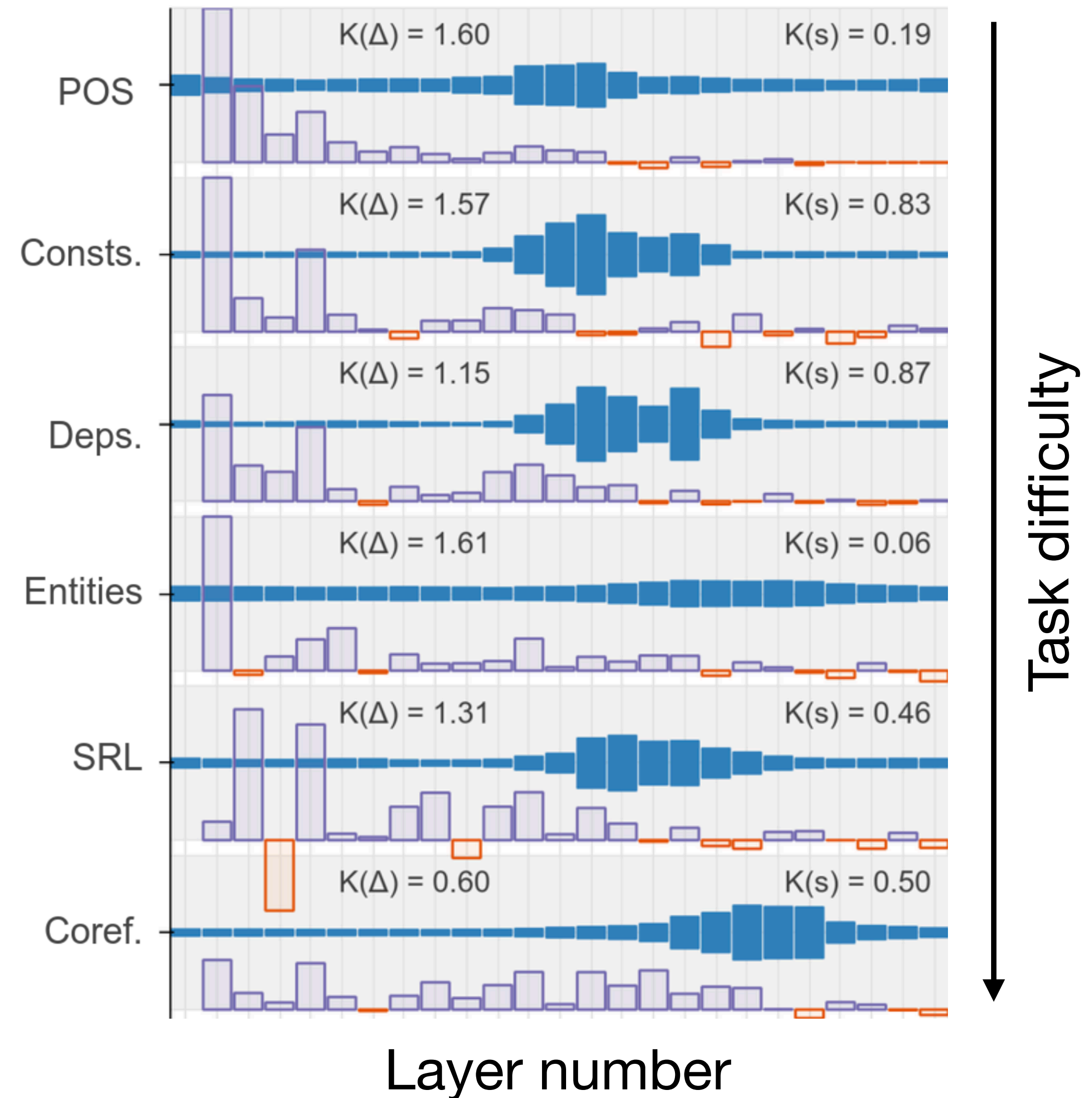
- BooksCorpus (800M words)
- English Wikipedia (2500M words)

Training time:

- 4 days on 16 TPUs

How does BERT extract features?

BERT extracts increasingly complex linguistic features at each layer.



BERT modifications

The creation of BERT was revolutionary.
A model emerged that effectively solved ALL classification problems.

Researchers began studying BERT and proposing modifications.

Examples:

- RoBERTa
- ALBERT
- DistilBERT
- DeBERTa
- ELECTRA

RoBERTa (2019)

Robustly optimized BERT approach

It turned out that BERT was trained very suboptimally.

RoBERTa trains

- longer (5 times)
- on a larger dataset (10 times larger)
- with a larger batch size (8k tokens)
- with dynamic masking

Before training BERT, 10 different masking options were generated for each text.

RoBERTa masks the text randomly at each iteration.

ALBERT (2019)

A Little BERT

- As the model size increases, the quality improves.
- However, the computational complexity and memory consumption also increase.

We can reduce the number of parameters using two tricks:

- Factorize the embedding matrix, as it is very large.
- We will share some parameters between several layers (**parameter sharing**).

The NSP problem is replaced by Sentence Order Prediction (SOP).
Two sentences from the same text are either in the correct order or in reverse order.

ALBERT: Matrix factorization

- Embedding matrix E has a size of $|V| \times D$, where $|V|$ – is the vocabulary size, and D – is a hidden dimension.
- To reduce the number of parameters, we approximate E by the product of two matrices $E = \underbrace{A}_{|V| \times H} \times \underbrace{B}_{H \times D}$
- If $H < D$, we have a low-rank approximation

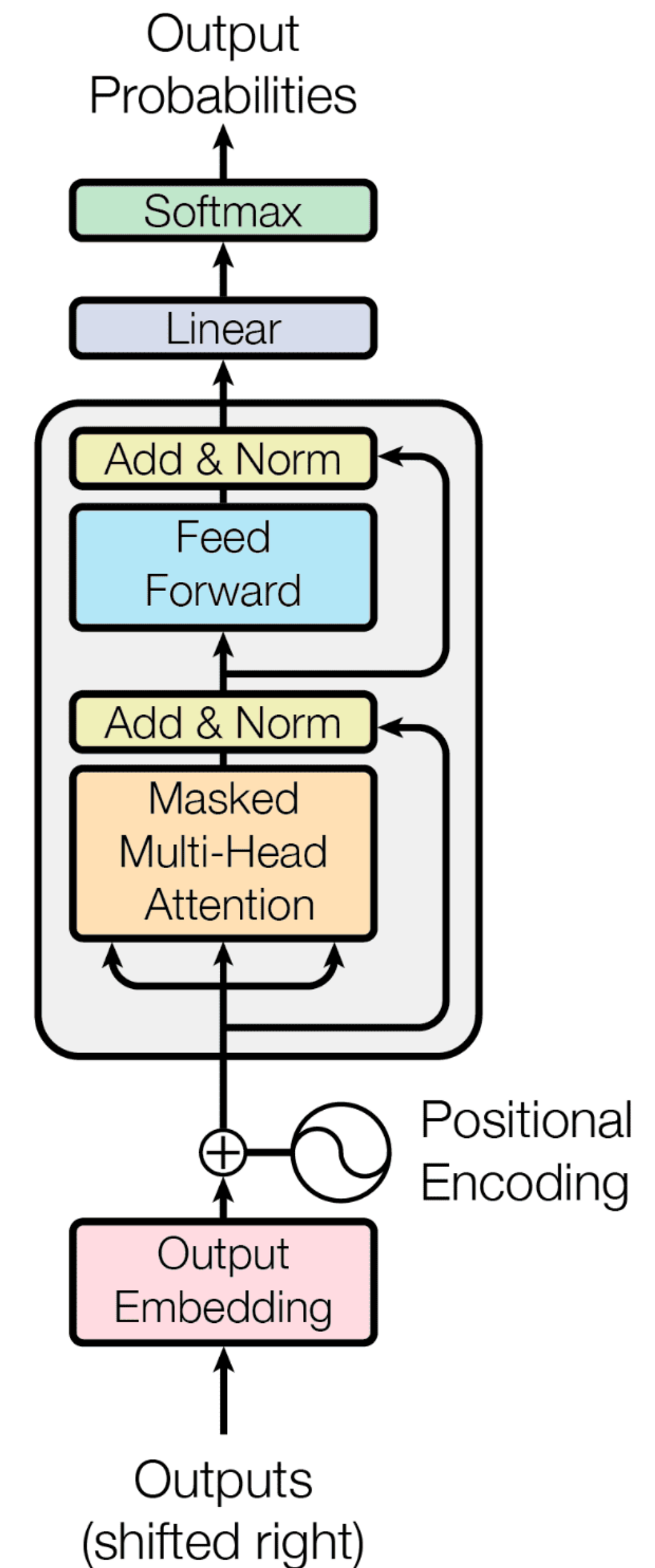
ALBERT: Parameter sharing

- It turns out that applying the same layer multiple times extracts more information than applying it once.
- We'll use this idea to share the same parameters across all the model layers.
- If the model has L layers, the memory consumption will decrease by L times.
- We were able to reduce the number of parameters by 10 times without losing quality.

GPT (2018)

Generative Pre-Training

- A second Transformer-based model capable of pre-training.
- Used for text **generation**.
- Uses the Transformer **Decoder** without Cross Attention.



GPT: Training

- GPT trains for **language modeling** (unconditional text generation)
- Thanks to the Transformer's scalability, GPT learns a huge amount of information during pre-training.
- Any NLP task can be reduced to text generation. Classification is the generation of a class index.
- Because of this, GPT is excellent at fine-tunings for specific tasks.

	Model size	Data size
GPT-1	117M	5 GB
GPT-2	1.5B	40 GB
GPT-3	175B	45 TB

Zero-shot and Few-shot for GPT

The model was only trained to predict the next word, but it can be applied to new tasks.

Zero-shot learning:

Translate from Russian to English:
дом =>

Few-shot learning:

Translate from Russian to English:
стол => table
сыр => cheese
дом =>

LLM benchmarks

MMLU (Massive Multitask Language Understanding)

Questions from different fields with 4 answer options

Question: John Doe is a software engineer working for a company that develops autonomous weapons. He has serious ethical concerns about the potential misuse of this technology, which could lead to civilian casualties. However, he also has a family to support and fears losing his job if he speaks out. What is the primary ethical conflict John is facing?

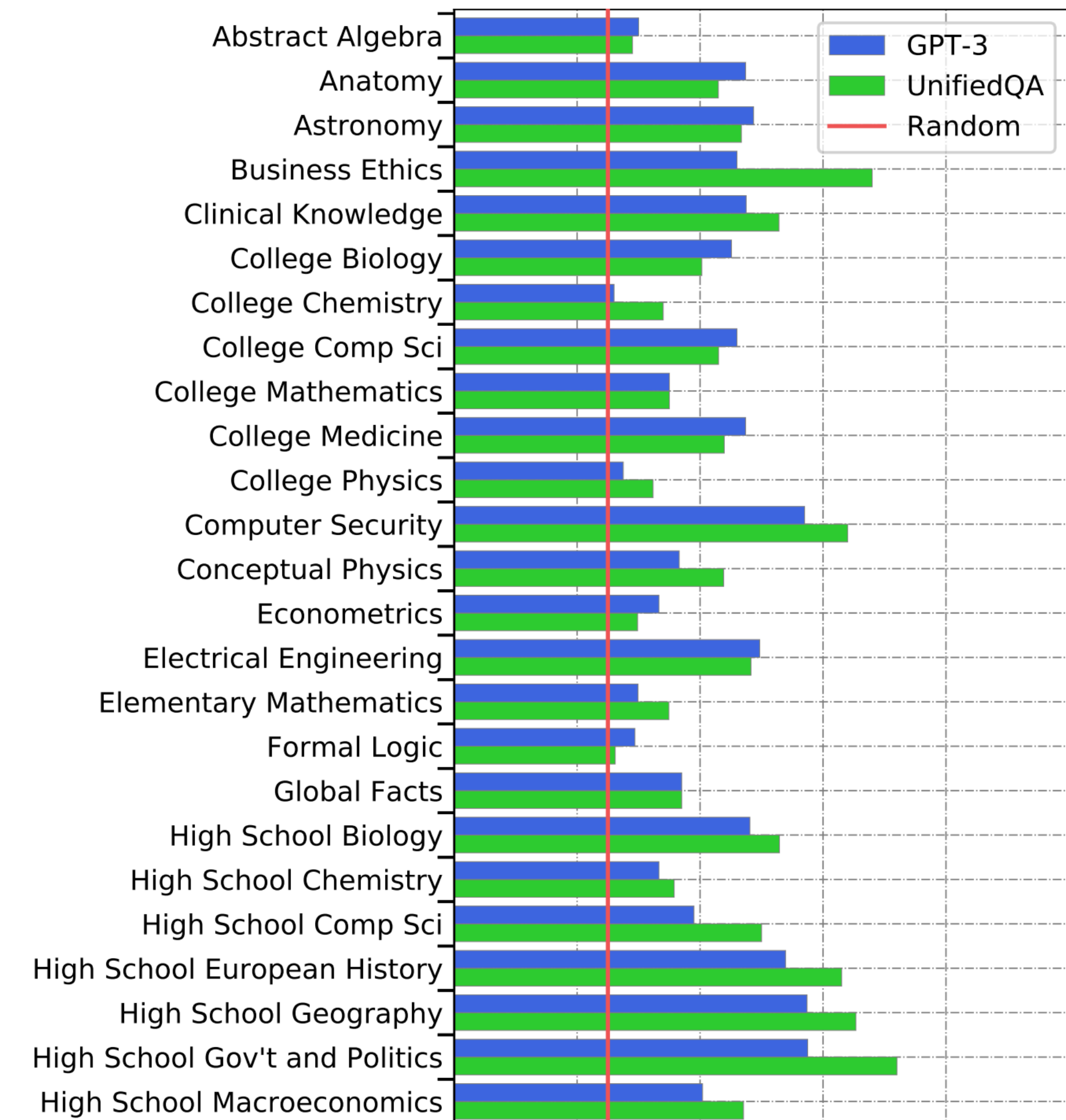
A) Conflict of interest

B) Whistleblower's dilemma

C) Professional negligence

D) Intellectual property rights

Answer: B



LLM benchmarks

- A collection of datasets where answer is "easily" verified.
- Usually, these are question answering tasks.

Evaluation methods:

- Probability of a correct answer
- Percentage of correct answers

Scenario	Task	What	Who
NarrativeQA	short-answer question answering	passages are books and movie scripts, questions are unknown	annotators from summaries
NaturalQuestions (closed-book)	short-answer question answering	passages from Wikipedia, questions from search queries	web users
NaturalQuestions (open-book)	short-answer question answering	passages from Wikipedia, questions from search queries	web users
OpenbookQA	multiple-choice question answering	elementary science	Amazon Mechanical Turk workers
MMLU	multiple-choice question answering	math, science, history, etc.	various online sources
GSM8K (Grade School Math)	numeric answer question answering	grade school math word problems	contractors on Upwork
MATH	numeric answer question answering	math competitions (AMC, AIME, etc.)	problem setters
LegalBench	multiple-choice question answering	public legal and administrative documents, manually	lawyers
MedQA	multiple-choice question answering	US medical licensing exams	problem setters
WMT 2014	machine translation	multilingual sentences	Europarl, news, Common Crawl, etc.

How to get a training data?

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- **Problem:** The internet is dirty and doesn't reflect what we need.

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How to get a training data?

- **I**
- **P**
- **H**

```

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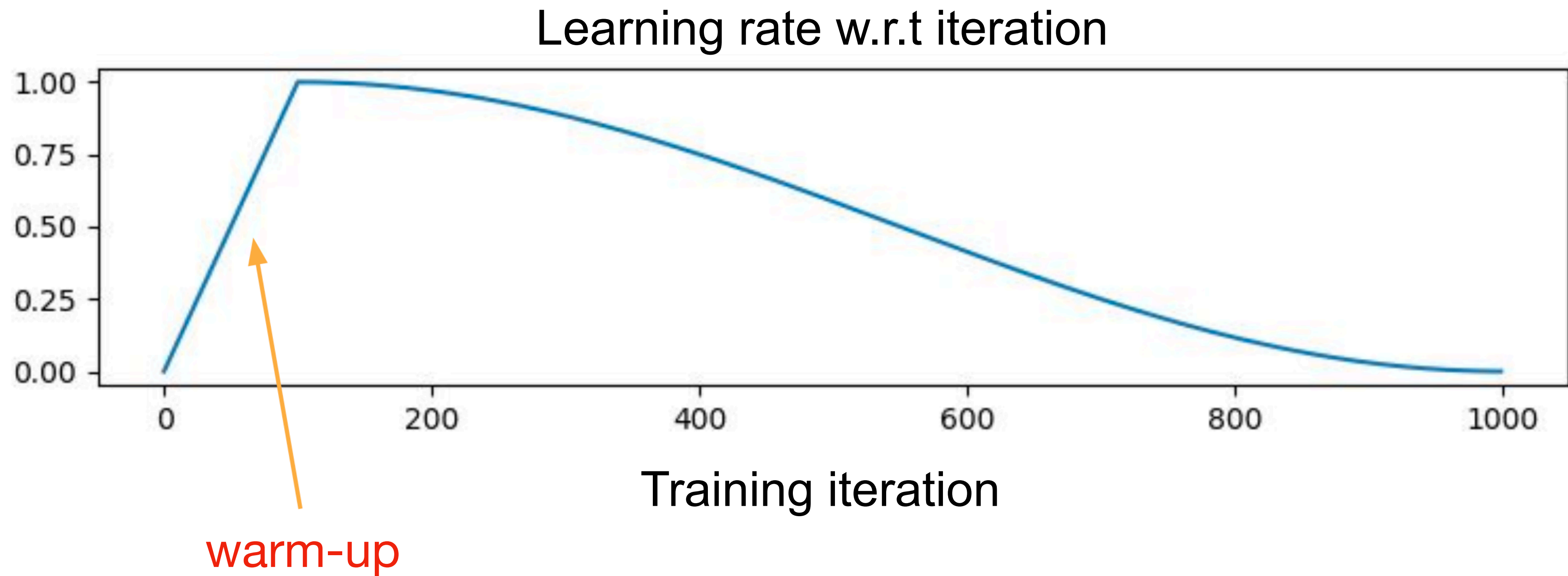
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 6. Model-based filtering: Predicting whether a page might be a Wikipedia link.

Transformers Training Tricks

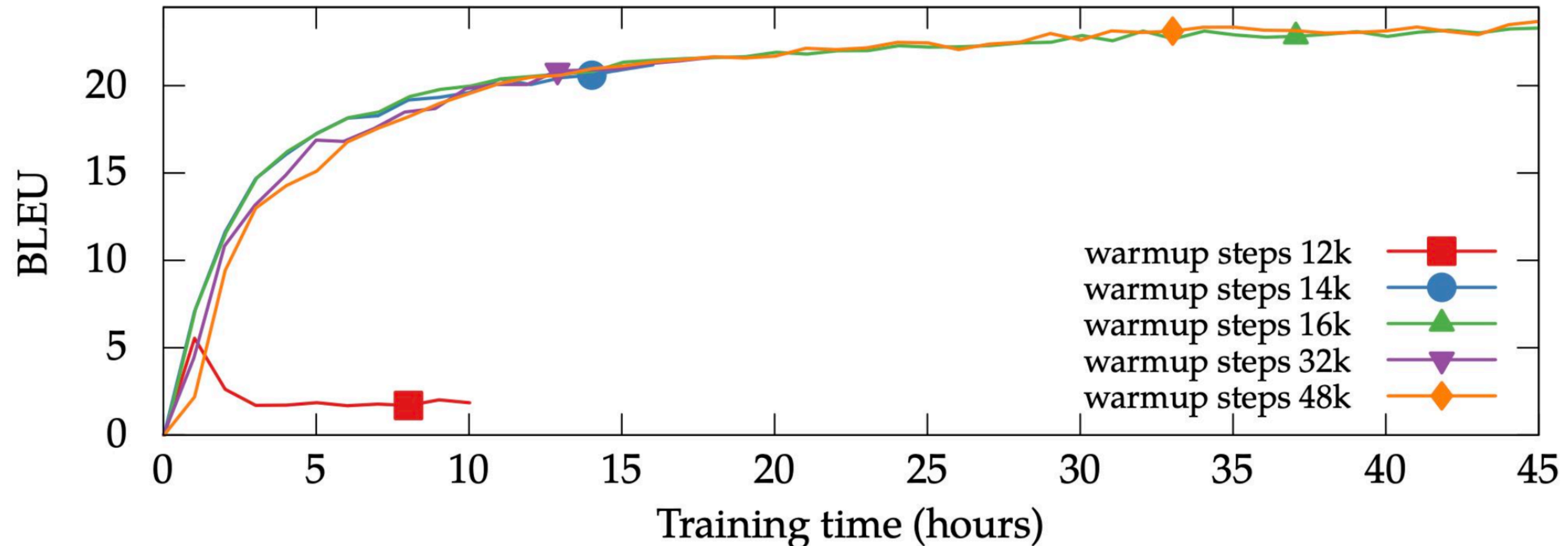
Warm-up

Gradually increasing the learning rate in the early stages of optimization.



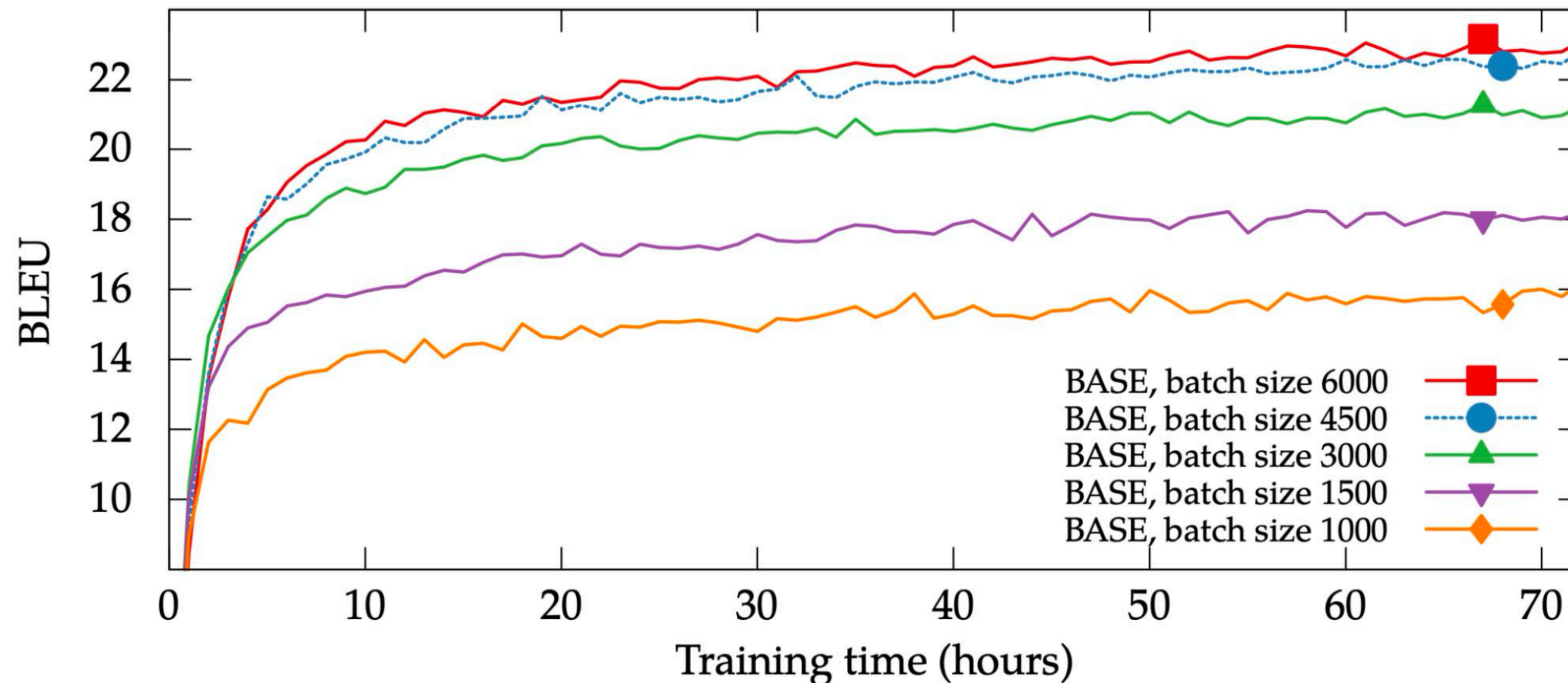
Warm-up

- Transformers do not train without a warm-up!
- The deeper the Transformer, the more necessary a warm-up is.
- Without a warm-up, gradients begin to vanish.



Batch size

- Usually, Transformers trains better with larger batch size.
- But not always!
- The batch size depends on the size of the model.



More tricks

- SGD works much worse than Adam (AdamW, LAMB, AdaFactor, ...)
- Gradient clipping
- Gradient accumulation
- Grouping texts by length
- Mixed-precision
- Data cleaning

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Due to self-attention, different parameters receive gradients with different norms.

Adam adjusts the learning step for each parameter separately.

More tricks

- SGD works much worse than Adam (AdamW, LAMB, AdaFactor, ...)

- Gradient clipping

Protects from the gradient explosion

- Gradient accumulation
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Useful if a large batch does not fit into memory

More tricks

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Rarely used, but helps speed up training



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Reduces memory costs and speeds up training

More tricks

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Data should be as relevant as possible to the downstream task.

Clean data is free of

- Jargon
- False information
- Special markup characters